5 (a)(i)	$V_X = \frac{R_X}{R_X+R_{TH}}E$ $P_X = V_X^2 / R_X$ $= (\frac{R_X}{R_X + R_{TH}}E)^2 / R_X$ $= \underline{\frac{R_X}{(R_X + R_{TH})^2}E^2}$	**C1** **A1**
(ii)	When temperature decreases, **resistance of thermistor R_{TH} increases**. This means that overall current in the **circuit will decrease**. OR **Using the expression in (a)(i), since R_{TH} increases,** Hence, **P_X decreases** since $P_X = I^2R_X$.	**M1** **A1**
(iii) (A)	At balanced potential (when there is no current detected), it means that $E_S = V_{TH}$. $E_S = V_{TH}$ $= \frac{R_{TH}}{R_X + R_{TH}}E$ $= \frac{2R_X}{R_X + 2R_X}E$ $= \frac{2}{3}E = 0.67E$	 **C1** **A1**
(iii) (B)	When X is replaced with a resistor of lower resistance, it means that V_{TH} will increase. **V_{TH} will be greater than E_S**. This will result in a **current flowing** in the secondary circuit, which will be **detected by the galvanometer**.	**B1** **B1**
(b)(i)	$R = pl / A$ $= (4.9 \times 10^{-7})(0.800) / [(\pi/4)(0.027/100)^2]$ $= 6.8\ \Omega$	 **M1**
(ii) (A)	$R_{eff} = 3.4 + (1/1.7 + 1/6.8)^{-1}$ $= 4.76\ \Omega$ Current $= 6.0 / 4.76$ $= 1.26$ $= \underline{1.3\ A}$ (2 sf)	**C1** **A1**
(ii) (B)	Voltmeter reading = $\underline{6.0\ V}$ (equal to e.m.f. of circuit)	**A1**